

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN
COURSE CODE: CSA1104

COURSE OUTCOME COVERED

***CO1:** Apply object-oriented concepts and software development life cycle models to design structured and modular software systems. (BL2, BL3)*

Assessment Tool Weightage: 60%

Dr Gayathri A

QUESTIONS: 12

Total Marks:60

Annexure A

CONCEPT MAPPING

Code of Conduct:

*I **Durga Sree R / 192411189** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

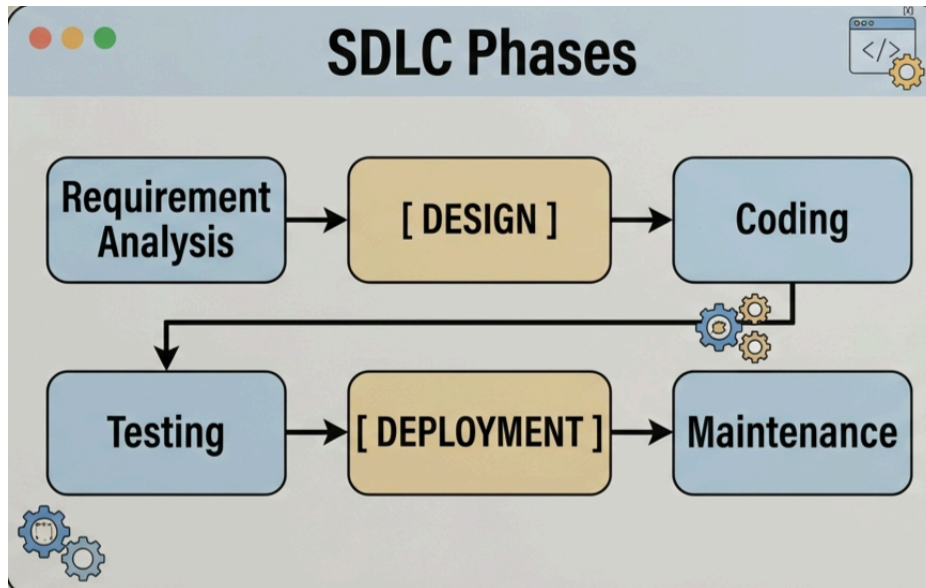
Signature of the student:

CONCEPT MAPPING

1. Construct a Concept Map

Given:

Requirement Analysis → _____ → Coding → Testing → _____ → Maintenance



Missing Concepts:

- Design
- Deployment

Explanation:

Requirement Analysis identifies the user's needs and system requirements. Design converts these requirements into software architecture. Coding converts the design into executable code. Testing verifies that the software functions correctly. Deployment releases the software to users. Maintenance provides updates and bug fixes after release.

Analysis:

- Ensures a systematic Software Development Life Cycle (SDLC).
- Minimizes development risks.
- Improves software quality and maintainability.

2. Construct a Concept Map

Given:

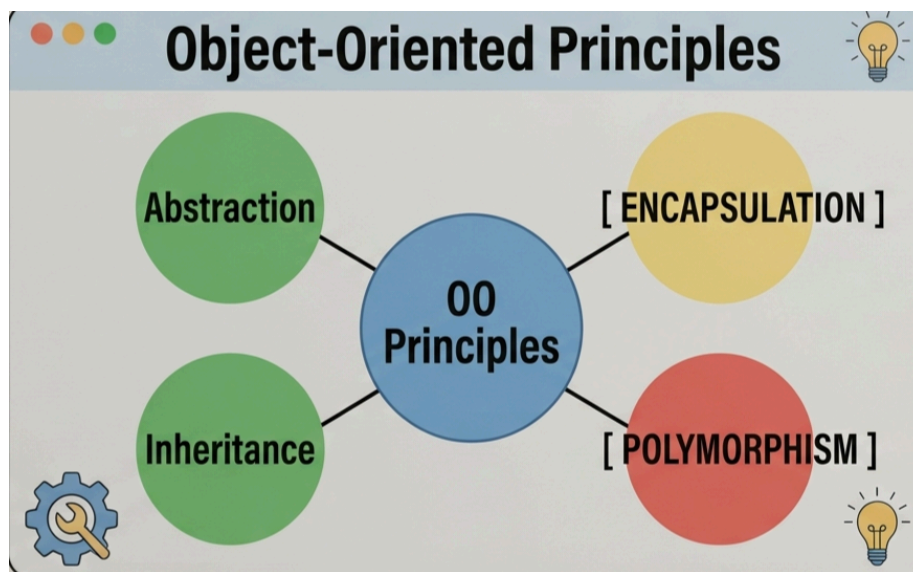
Object-Oriented Principles

→ Abstraction

→ _____

→ Inheritance

→ _____



The missing concepts are:

- Encapsulation
- Polymorphism

Explanation:

Abstraction hides unnecessary implementation details. Encapsulation combines data and methods and protects data from direct access. Inheritance allows a class to acquire properties of another class. Polymorphism allows the same interface or method to have different behaviors.

Analysis:

- Improves software security.
- Supports code reusability.
- Provides flexibility in software design.

3. Construct a Concept Map

Given:

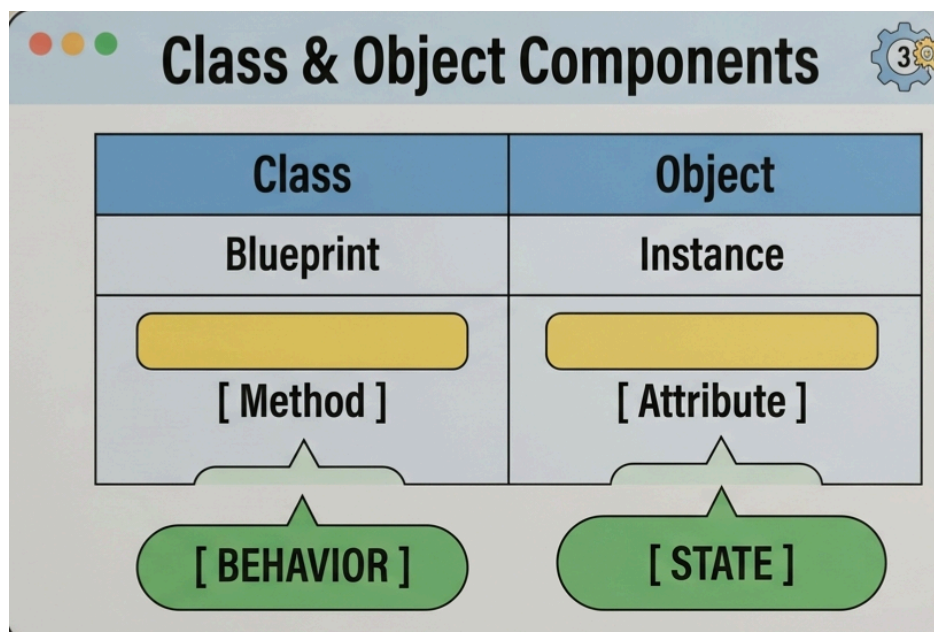
Class → Blueprint

Object → Instance

Method → _____

Attribute → _____

Answer:



The missing concepts are:

- Behavior
- State

Explanation:

A class acts as a blueprint for creating objects. An object is an instance of a class. A method defines the behavior or actions of an object. An attribute represents the state or properties of an object.

Analysis:

- Helps represent real-world entities.
- Defines object properties and actions.
- Improves system organization.

4. A Concept Map Shows

Given:

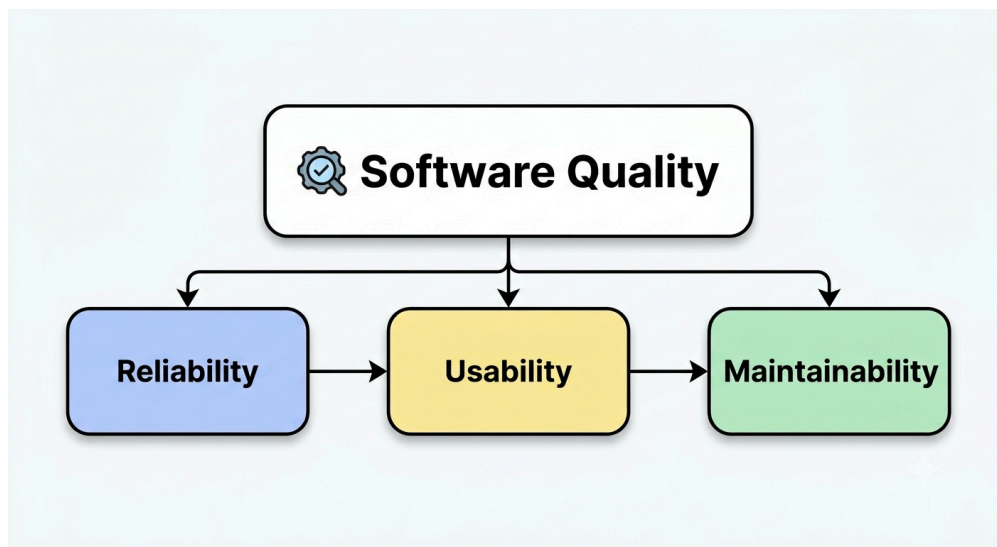
Software Quality

→ Reliability

→ _____

→ Maintainability

Answer:



The missing concept is:

- Usability

Explanation:

Reliability ensures that software performs correctly and consistently. Usability measures how easily users can understand and operate the software. Maintainability makes the software easy to modify, update and repair.

Analysis:

- Improves user satisfaction.
- Increases software acceptance.
- Enhances overall software quality.

5. UML Concept Map

Given:

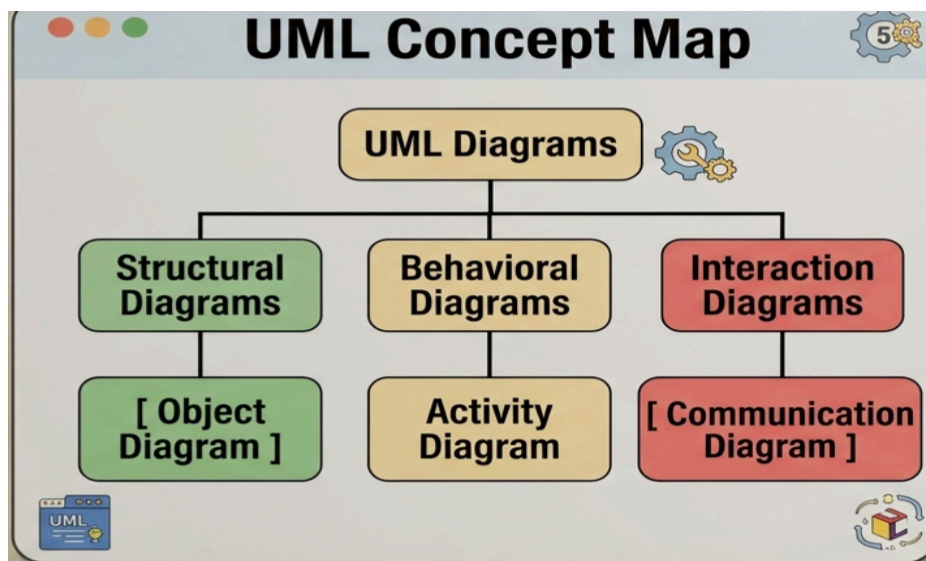
UML

→ Structural Diagrams → _____

→ Behavioral Diagrams → Activity Diagram

→ Interaction Diagrams → _____

Answer:



The missing concepts are:

- Object Diagram
- Communication Diagram

Explanation:

Object Diagrams represent object instances and their relationships at a particular time. Activity Diagrams represent the workflow of activities. Communication Diagrams show how objects exchange messages with each other.

Analysis:

- Improves system visualization
- Supports object-oriented analysis and design.
- Helps communicate system behavior clearly.

6. Given:

Requirements ✓ → Design ✓ → Coding ✗ → Testing ✗ → Deployment ✗

Explanation:

The failure occurs during the Coding phase. The implementation may contain syntax errors, logical errors or incorrect program logic. Because coding is incomplete or incorrect, testing and deployment cannot proceed successfully.

Analysis:

- Errors affect the subsequent phases.
- Testing cannot be performed effectively.
- Software quality may decrease.

Solution:

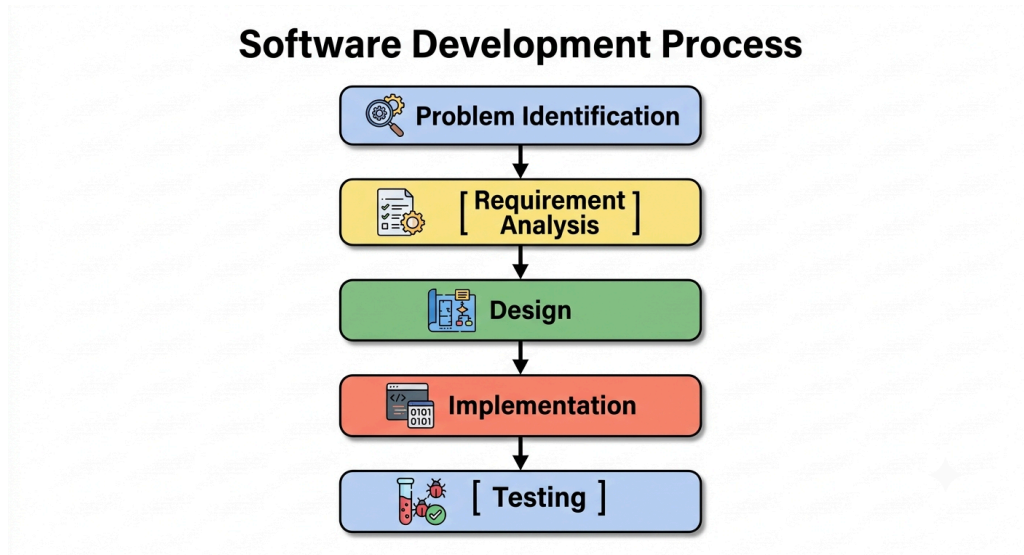
- Review and correct the source code.
- Follow coding standards.
- Perform code inspection before testing.

7. Construct a Concept Map

Given:

Problem Identification → _____ → Design → Implementation → _____

Answer:



The missing concepts are:

- Requirement Analysis
- Testing

Explanation:

Requirement Analysis identifies the user needs and system requirements. Design converts these requirements into a system structure. Implementation develops the software. Testing verifies whether the implemented system satisfies the specified requirements.

Analysis:

- Reduces development errors.
- Improves software quality.
- Ensures successful project completion.

8. Construct a Concept Map

Given:

Object-Oriented Features

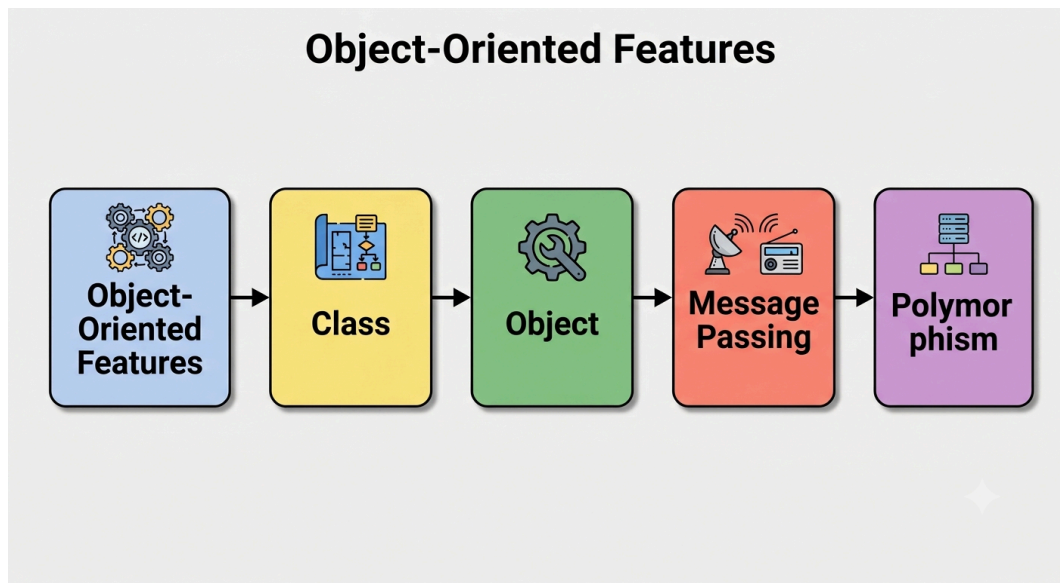
→ Class

→ _____

→ Message Passing

→ _____

Answer:



The missing concepts are:

- Object
- Dynamic Binding

Explanation:

A class is a blueprint used to create objects. An object is an instance of a class. Message Passing allows objects to communicate by calling methods. Dynamic Binding determines the method to be executed at runtime.

Analysis:

- Supports runtime polymorphism.
- Provides flexibility in software design.
- Improves code maintainability.

9. Construct a Concept Map

Given:

Software Development Models

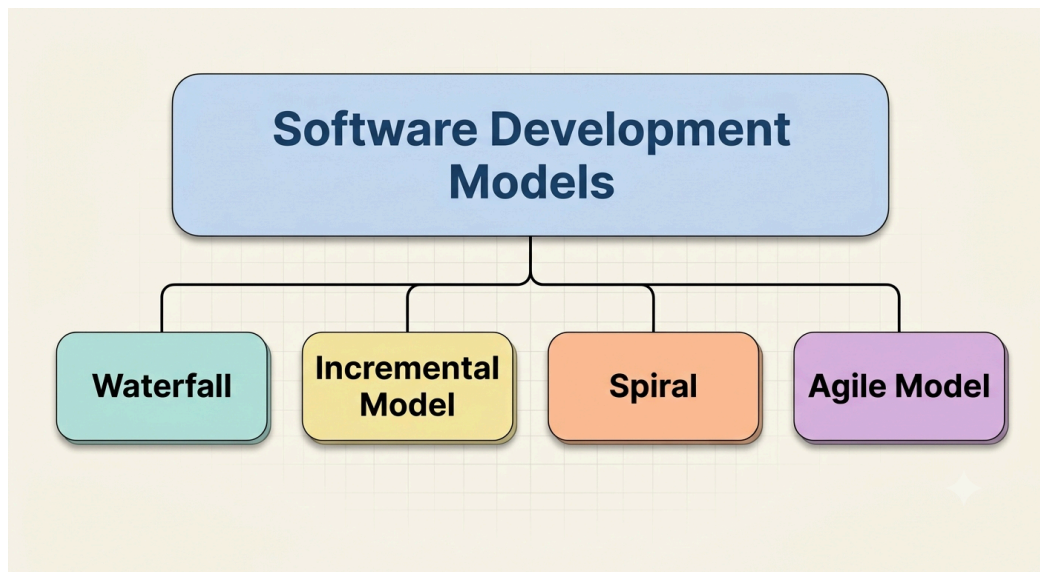
→ Waterfall

→ _____

→ Spiral

→ _____

Answer:



The missing concepts are:

- Incremental Model
- Agile Model

Explanation:

The Waterfall Model follows sequential development phases. The Incremental Model develops and delivers software in small increments. The Spiral Model focuses on iterative development and risk analysis. Agile Model supports iterative development with continuous customer feedback.

Analysis:

- Improves adaptability.
- Reduces development risks.
- Increases customer satisfaction.

10. Construct a Concept Map

Given:

Class Diagram

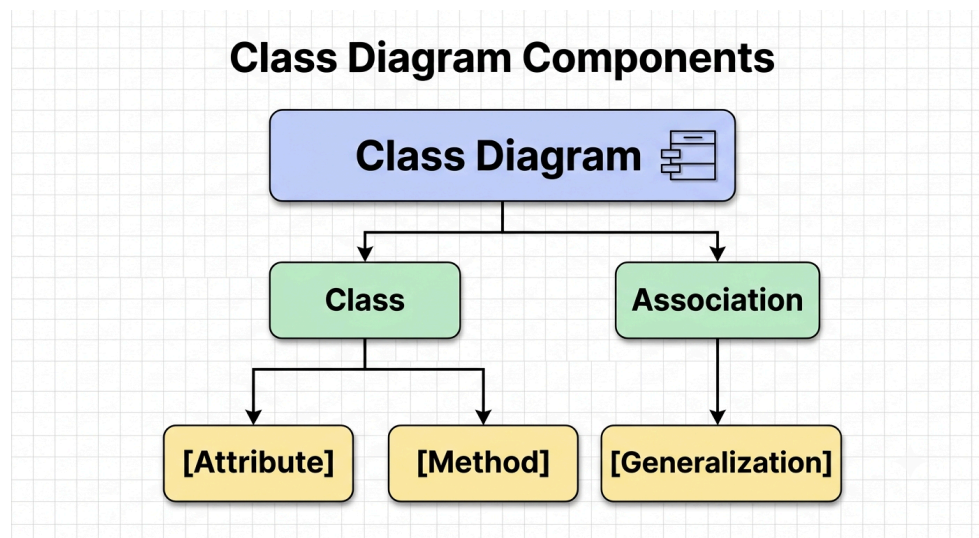
→ Class

→ _____

→ Association

→ _____

Answer:



The missing concepts are:

- Attribute
- Method

Explanation:

A Class represents a set of similar objects. Attributes define the properties or data of a class. Association represents the relationship between classes. Methods define the operations or functions performed by a class.

Analysis:

- Represents object structure clearly.
- Supports object-oriented design.
- Improves software documentation.

11. Construct a Concept Map

Given:

Software Engineering Principles

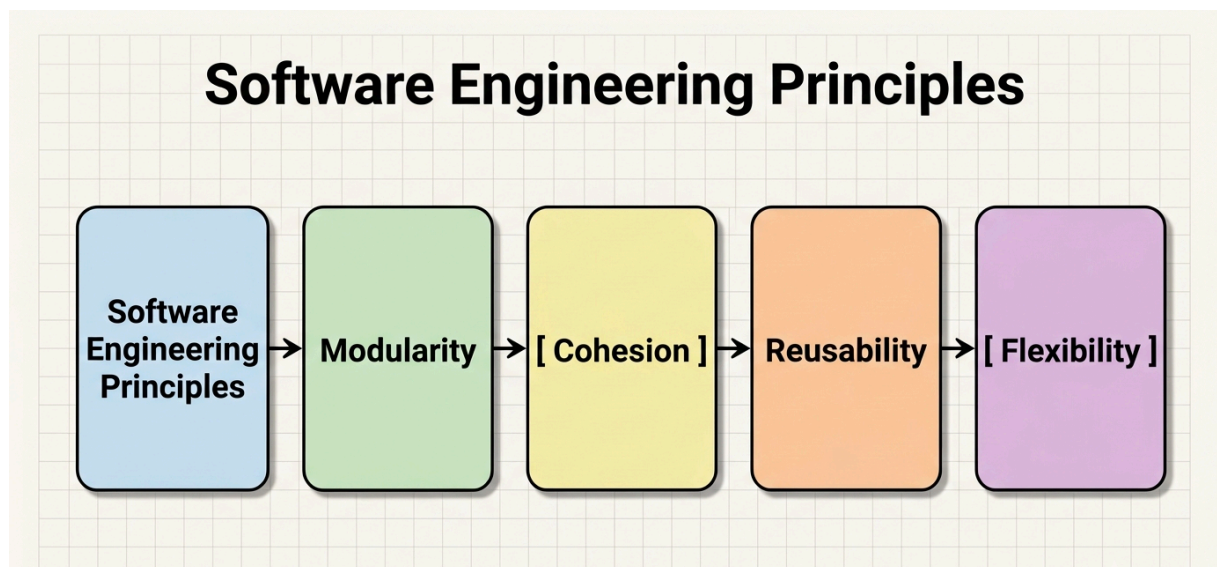
→ Modularity

→ _____

→ Reusability

→ _____

Answer:



The missing concepts are:

- Maintainability
- Scalability

Explanation:

Modularity divides a system into smaller independent modules. Maintainability makes the software easier to modify and update. Reusability allows existing components to be used again. Scalability allows the system to handle increasing workload.

Analysis:

- Reduces maintenance effort.
- Supports code reuse.
- Helps long-term software growth.

12. Given:

Requirement Analysis ✓ → System Design ✓ → Coding ✓ → Integration ✗ → Deployment ✗

Explanation:

The failure occurs during the Integration phase. Individual modules may not communicate correctly due to interface or compatibility problems. This prevents the complete system from working properly and delays deployment.

Analysis:

- Causes functional inconsistencies.
- Increases debugging effort.
- Delays software deployment.

Solution:

- Perform integration testing.
- Verify module interfaces.
- Resolve compatibility issues before deployment.

CONCEPT MAPPING

RUBRICS FOR CALCULATION

CONCEPT MAPPING

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand